Strand 3: Geometry and Measurement

Sub-strand 2: Measurement

Lesson 1: Circles, Semicircles, Scale Drawings, and Measurement

1. Curriculum Standard and Overview

- **Strand**: 3 — Geometry and Measurement
- **Sub-strand**: 2 — Measurement
- **Content Standard**: B9.3.2.1 (measurement of plane figures and scale)
- **Indicators**: B9.3.2.1.1–1.3 (surface area of solids — partly covered in later lessons), with circle/scale measurement as the mock-tested core here.
- **Mock-paper links**: P1 Q5 (area $64\pi \rightarrow$ circumference), P1 Q24 (scale 20 m $\rightarrow$ 10 cm), P2 Q3(d) (area of semicircle, $r = 7$ cm).

2. Preamble — Measuring Round Things

Circles are not polygons. We measure them with two special numbers:

- **Circumference** — the distance **around** a circle.
- **Area** — the space **inside** a circle.

Both need π (pi). In BECE, use $\pi = (22)/(7)$ or 3.142 unless told otherwise (exact 64π style answers leave π in).

3. Circle Formulae

Quantity	Formula
Circumference	$C = 2\pi r$ or $C = \pi d$
Area	$A = \pi r^2$
Semicircle area	$(1)/(2)\pi r^2$
Semicircle perimeter	$\pi r + 2r$ (curved part + diameter)

Working backwards: if $A = 64\pi$, then $\pi r^2 = 64\pi \Rightarrow r^2 = 64 \Rightarrow r = 8$. Then $C = 2\pi r = 16\pi$.

Mock P1 Q5: Area of a circle is 64π. Circumference? $r = 8$, so $C = 16\pi$. **Answer: B — 16π.**

Mock P2 Q3(d): Area of a semicircle with $r = 7$ cm, $\pi = (22)/(7)$. $A = (1)/(2)\pi r^2 = (1)/(2) \times (22)/(7) \times 49 = (1)/(2) \times 154 = 77\ cm^2$

Answer: 77 cm².

4. Scale Drawings

A **scale** compares a drawing length to the real length.

4.1 Writing a scale

Scale = $(drawing)/(real)$ (same units).

Mock P1 Q24: Hall 20 m long drawn as 10 cm. Scale?

$(10\ \text{cm})/(20\ \text{m}) = (10\ \text{cm})/(2000\ \text{cm}) = (1)/(200)$

Answer: A — 1 : 200.

4.2 Using a scale

If scale is $1 : 50\,000$, then 4 cm on the map = $4 \times 50\,000 = 200\,000$ cm = 2 km.

4.3 Real → drawing

A 30 m room drawn at $1 : 500$: 30 m = 3000 cm; drawing = $3000 \div 500 = 6$ cm.

5. Other Measurement Essentials

- 1 m = 100 cm; 1 km = 1000 m = $100\,000$ cm.
- 1 litre = 1000 cm³; 1 m³ = 1000 litres.
- Time: 1 day = 24 h; 1 h = 60 min; 1 min = 60 s.

6. Twenty Fully Worked Examples

Example 1 — Mock P1 Q5

$A = 64\pi \Rightarrow r = 8 \Rightarrow C = 16\pi$. **B.**

Example 2 — Mock P2 Q3(d)

Semicircle $r = 7$, $\pi = 22/7$: $A = 77$ cm².

Example 3 — Mock P1 Q24

20 m → 10 cm ⇒ scale 1 : 200. **A.**

Example 4 — Circumference from radius

$r = 14$ cm, $\pi = 22/7$: $C = 2 \times 22/7 \times 14 = 88$ cm.

Example 5 — Area from radius

$r = 7$ cm: $A = 22/7 \times 49 = 154$ cm².

Example 6 — Radius from area

$A = 154$ cm², $\pi = 22/7$: $r^2 = 49 \Rightarrow r = 7$ cm.

Example 7 — Diameter given

$d = 20$ cm ⇒ $r = 10$; $C = \pi d = 20\pi$ cm.

Example 8 — Semicircle perimeter

$r = 7$: $P = \pi r + 2r = 22 + 14 = 36$ cm.

Example 9 — Scale: map to ground

Scale 1 : 50 000, map 4 cm ⇒ 2 km.

Example 10 — Scale: ground to map

30 m at 1 : 500 ⇒ 6 cm.

Example 11 — Unit conversion

3 m = 300 cm.

Example 12 — Unit conversion

2.5 km = 2500 m.

Example 13 — Time conversion

31 days = 44 640 minutes.

Example 14 — Area of rectangle

$l = 8$ cm, $w = 5$ cm ⇒ $A = 40$ cm².

Example 15 — Perimeter of rectangle

$l = 8$ cm, $w = 5$ cm ⇒ $P = 26$ cm.

Example 16 — Triangle area

$b = 10$ cm, $h = 6$ cm ⇒ $A = 30$ cm².

Example 17 — Trapezium area (preview)

$a = 8, b = 5, h = 4$: $A = (1)/(2)(a+b)h = 26$.

Example 18 — Circle from diameter

$d = 10$ cm ⇒ $A = \pi\ (5)^2 = 25\pi$ cm².

Example 19 — Compare circle areas

r doubles ⇒ area × 4.

Example 20 — Composite shape

Rectangle 20 × 10 with semicircle (r = 5) on top: $A = 200 + (1)/(2)\pi(25) = 200 + 12.5\pi$.

7. Comprehensive Exercise Sets (40 Questions)

All 3 authentic mock items for this topic are included. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Circumference & area (includes mock)

1. **[Mock 2026 P1 Q5]** If the area of a circle is 64π, what is the circumference?

A. 8π B. 16π C. 32π D. 64π

2. Find the circumference of a circle with $r = 14$ cm ($\pi = 22/7$).

3. Find the area of a circle with $r = 7$ cm ($\pi = 22/7$).
4. The area of a circle is 154 cm². Find its radius ($\pi = 22/7$).
5. **Spiral review:** Find $12x + 9q$ when $x = 11$, $q = 3$.

Exercise 2 — Semicircles (includes mock)

1. **[Mock 2026 P2 Q3(d)]** Find the area of a semi-circle whose radius is 7 cm. [Take $\pi = 22/7$]
2. Find the perimeter of a semicircle with $r = 7$ cm (straight edge = diameter).
3. Find the area of a semicircle with $r = 14$ cm ($\pi = 22/7$).
4. **Spiral review:** $2x + 1 > 9$. Solve.
5. **Spiral review:** $13x - 12 = 5x + 60$. Find x.

Exercise 3 — Scale drawings (includes mock)

1. **[Mock 2026 P1 Q24]** A hall 20 m long is represented on a diagram as 10 cm. What is the scale?

A. 1 : 200 B. 1 : 250 C. 1 : 400 D. 1 : 500

2. A map has scale 1 : 50 000. What distance does 4 cm represent in km?
3. A 30 m room is drawn at scale 1 : 500. How long is the drawing?
4. **Spiral review:** $20xy - 5xy = ?$
5. **Spiral review:** Classify -3 under $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}$.

Exercise 4 — Unit conversion

1. 3 m = ? cm
2. 2.5 km = ? m
3. 1 litre = ? cm³
4. 31 days = ? minutes (leave in ordinary form)
5. **Spiral review:** $(-3)^{(x-2)} = -243$. Find x.

Exercise 5 — Rectangles & triangles

1. Find the area of a rectangle 8 cm by 5 cm.
2. Find the perimeter of a rectangle 8 cm by 5 cm.
3. Find the area of a triangle with base 10 cm and height 6 cm.
4. A rectangle has perimeter 30 cm and length 8 cm. Find the width.
5. **Spiral review:** Evaluate $(2)/(3) \times (1)/(4) + (2)/(5) \div (14)/(15)$.

Exercise 6 — Circles from diameter

1. $d = 20$ cm. Find the circumference in terms of π.
2. $d = 10$ cm. Find the area in terms of π.
3. If r doubles, what happens to the area?
4. **Spiral review:** Find $12x + 9q$ when $x = 11$, $q = 3$.
5. **Spiral review:** A van travels 154 km in $1(3)/(4)$ h. Speed?

Exercise 7 — Working backwards

1. The area of a circle is 154 cm². Find r.
2. The circumference of a circle is 44 cm. Find r ($\pi = 22/7$).
3. The area of a semicircle is 77 cm². Find r ($\pi = 22/7$).
4. **Spiral review:** Solve $(4)/(3x) = 7 - (3)/(x)$.
5. **Spiral review:** Order $(3)/(8), (1)/(2), (5)/(16)$.

Exercise 8 — Composite & challenge

1. A rectangle 20 by 10 has a semicircle of radius 5 on top. Find the total area in terms of π.

2. A circle has area 36π. Find its circumference.
3. A square of side 14 cm has a circle of radius 7 cm cut out. Find the remaining area ($\pi = 22/7$).
4. **Spiral review:** Three consecutive odd integers sum to 33. Find them.
5. **Spiral review:** Find the gradient through $(1, 2)$ and $(4, -4)$.

8. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. $r = 8$, $C = 16\pi \rightarrow$ **B**.
2. $C = 2 \times 22/7 \times 14 = 88$ cm.
3. $A = 22/7 \times 49 = 154$ cm².
4. $r^2 = 49 \Rightarrow r = 7$ cm.
5. 159.

Solutions to Exercise 2

1. $A = (1)/(2) \times 22/7 \times 49 = 77$ cm².
2. $P = 22 + 14 = 36$ cm.
3. $A = (1)/(2) \times 22/7 \times 196 = 308$ cm².
4. $x > 4$.
5. $x = 9$.

Solutions to Exercise 3

1. $10 : 2000 = 1 : 200 \rightarrow$ **A**.
2. $4 \times 50\,000 = 200\,000$ cm $= 2$ km.
3. $3000 \div 500 = 6$ cm.
4. $15xy$.
5. ■, ■.

Solutions to Exercise 4

1. 300 cm.
2. 2500 m.
3. 1000 cm³.
4. $44\,640$ minutes.
5. $x = 7$.

Solutions to Exercise 5

1. 40 cm².
2. 26 cm.
3. 30 cm².
4. $w = 7$ cm.
5. $(25)/(42)$.

Solutions to Exercise 6

1. $C = \pi\, d = 20\pi$ cm.
2. $A = 25\pi$ cm².
3. **Area × 4**.
4. 159.
5. 88 km/h.

Solutions to Exercise 7

1. $r = 7$ cm.
2. $2 \times 22/7 \times r = 44 \Rightarrow r = 7$ cm.
3. $(1)/(2) \times 22/7 \times r^2 = 77 \Rightarrow r^2 = 49 \Rightarrow r = 7$ cm.
4. $x = (13)/(21)$.
5. $(5)/(16) < (3)/(8) < (1)/(2)$.

Solutions to Exercise 8

1. $200 + 12.5\pi$.
2. $r = 6$, $C = 12\pi$.
3. $196 - 154 = 42$ cm².
4. **9, 11, 13**.
5. $m = -2$.

9. Authentic Mock Items in This Lesson

Ref	Question	Answer
P1 Q5	$A = 64\pi \rightarrow$ circumference	B — 16π
P2 Q3(d)	Semicircle $r = 7$	77 cm²
P1 Q24	20 m drawn as 10 cm	A — 1 : 200

10. Lesson Summary

Skill	Formula
Circumference	$C = 2\pi r = \pi d$
Area of circle	$A = \pi r^2$
Semicircle area	$(1)/(2)\pi r^2$
Scale	drawing : real (same units)
Conversion	1 m = 100 cm; 1 km = 1000 m

Lesson prepared under `Grade9-Maths/Rules.md` *— plain English, British notation, full workings required.*